ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 2

MECHATRONICS

PAPER 1 · PRACTICE G · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 2 Mechatronics paper 1 · practice g.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 2. It is not a Department of Higher Education and Training marking guideline.

How to mark: each ✓ is one mark. Give marks for the formula, the substitution and the answer. Accept other correct answers and wording. Symbols and circuits may be drawn in different but correct ways.

QUESTION 1

1.1 Class 1 – fulcrum between effort and load ✓, e.g. crowbar, seesaw, scissors ✓; class 2 – load between fulcrum and effort ✓, e.g. wheelbarrow, nutcracker ✓; class 3 – effort between fulcrum and load ✓, e.g. tweezers, fishing rod, human forearm ✓ (6)

1.2 $\text{MA} = \text{load} \div \text{effort} = 600 \div 150 \checkmark = 4 \checkmark$ (2)

1.3 Any TWO, ✓✓ each: no slipping, so a fixed speed ratio; can transmit larger forces; works in hot or wet conditions; long life. (4)

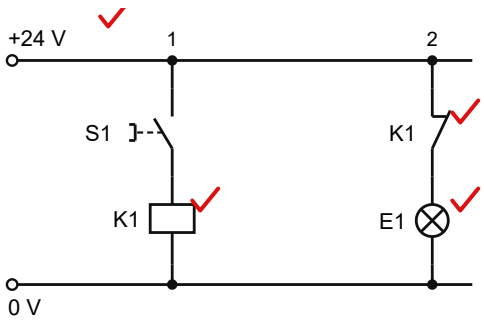
1.4 It supports a rotating shaft and reduces friction between moving parts ✓✓. Any TWO types ✓✓: ball bearing; roller bearing; plain (bush) bearing; needle bearing; thrust bearing. (4)

1.5 The effort must move 4 times as far as the load ✓✓; so (ignoring friction) the effort needed is about a quarter of the load ✓✓. (4)

[20]

QUESTION 2

2.1



Supply lines ✓; NC contact of K1 ✓ in series with lamp E1 ✓; correct symbols and labels ✓ (4)

2.2

A	B	C	Q
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

(all 8 combinations of A, B and C ✓✓; Q = 0 for the first seven rows ✓✓; Q = 1 only for 1 1 1 ✓) (5)

2.3 When START is pressed, the relay coil is energised ✓; a normally open contact of the same relay, connected in parallel with START, closes ✓; when START is released, current keeps flowing to the coil through this holding contact ✓, until a STOP button breaks the circuit ✓. (4)

2.4 A relay switches small currents in control circuits and signals ✓✓; a contactor is a heavy-duty relay built to switch large currents, such as motors, and often has auxiliary contacts and arc suppression ✓✓. (4)

2.5 Alternating current ✓; normally open ✓; normally closed ✓ (3)

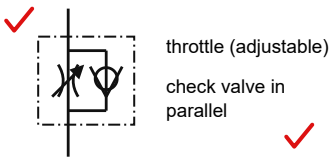
[20]

QUESTION 3

3.1 Any TWO ✓✓: reciprocating (piston) compressor; rotary screw compressor; rotary vane compressor; diaphragm compressor. (2)

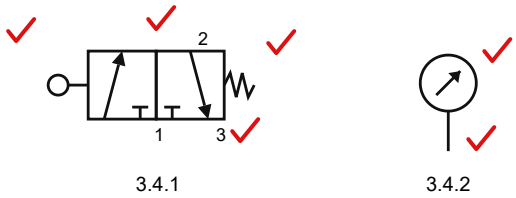
3.2 Water causes rust and corrosion inside pipes, valves and cylinders ✓; it washes away lubrication ✓; and it can freeze or cause valves to stick and fail ✓. (3)

3.3



With one-way flow control valves (throttle and check valve) ✓ fitted in the exhaust lines (meter-out) ✓; the throttle restricts the air leaving the cylinder, which sets the speed ✓; the check valve allows free flow into the cylinder ✓. Symbol: throttle with a check valve in parallel and an adjustment arrow ✓✓ (6)

3.4



3.4.1 3.4.2

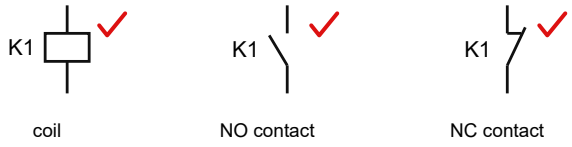
3.4.1 Two squares (2 positions) ✓, 3 ports correctly shown ✓, roller actuator ✓, spring return ✓. 3.4.2 Circle with a pointer (arrow) ✓ connected to the line ✓ (6)

3.5 It lets the air from a cylinder escape directly to the atmosphere close to the cylinder ✓✓, so the cylinder moves faster (it does not have to push the air back through the control valve) ✓. (3)

[20]

QUESTION 4

4.1

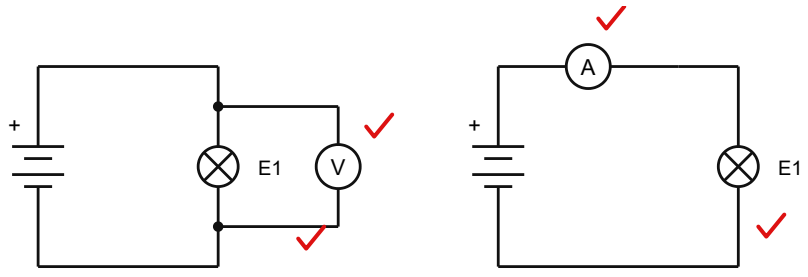


coil NO contact NC contact

Coil (rectangle) ✓; NO contact (open) ✓; NC contact (closed) ✓; labels K1 on all ✓ (4)

4.2 When current flows through the coil, it becomes an electromagnet ✓; it attracts the armature ✓, which moves the contacts: NO contacts close and NC contacts open ✓; when the current stops, a spring returns the contacts ✓. (4)

4.3



(a) voltmeter in parallel (b) ammeter in series

(a) Voltmeter range, probes across (in parallel with) the lamp ✓✓. (b) Ammeter range, the circuit is broken and the meter connected in series with the lamp ✓✓. (4)

4.4 The ohmmeter uses its own battery to measure ✓; the circuit's voltage gives wrong readings and can damage the meter ✓. (2)

4.5 Any THREE ✓✓✓: clean and quiet; easy to control speed and direction; efficient; available in many sizes; easy to connect to controllers; reliable with low maintenance. (3)

4.6 Live – brown (or red in old wiring) ✓; neutral – blue (or black) ✓; earth – green and yellow (or green) ✓ (3)

[20]

QUESTION 5

5.1 Any FOUR ✓✓✓✓: reduces friction; reduces wear; carries away heat; prevents rust and corrosion; keeps dirt out (sealing); reduces noise; saves energy. (4)

5.2 Any FOUR ✓✓✓✓: unusual noise (grinding, squealing); vibration; overheating; burning smell; drawing more current than normal; smoke; loose shaft (play); slower speed. (4)

5.3 Start at the input of the system (e.g. the supply or start button) ✓ and test each stage in turn ✓, moving towards the output ✓, until the stage where the signal is lost is found – that stage is faulty ✓. (4)

5.4 Any FOUR ✓✓✓✓: never point compressed air at a person; never use it to clean clothes or skin; wear safety glasses; release pressure before disconnecting hoses; check hoses and fittings for damage; keep pressure at the correct setting; secure hoses so they cannot whip. (4)

5.5 It shows what work was done and which parts were replaced ✓; it shows repeated faults so their cause can be found ✓; it helps to plan preventive maintenance and order spares ✓; it is a legal / safety record ✓. (4)

[20]

TOTAL: 100

END OF MARKING GUIDELINE